

Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

Depth-wise Coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} (f^{l+1}(X^l))_1 \end{aligned} \right\} A_{11}^{l,l+1}$$

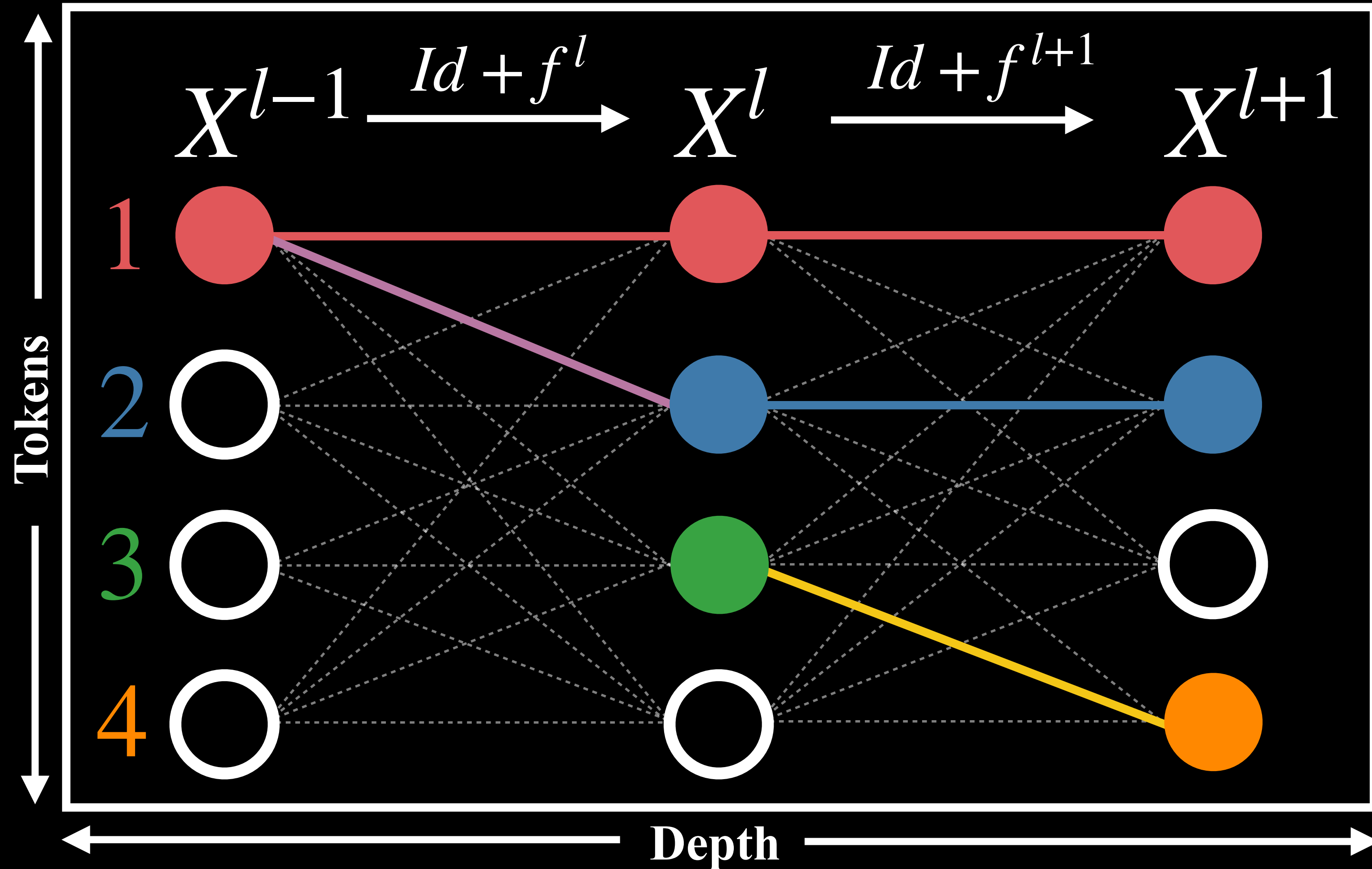
Token-wise Coupling

Self coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2 \end{aligned} \right\} A_{12}^{l,l+1}$$

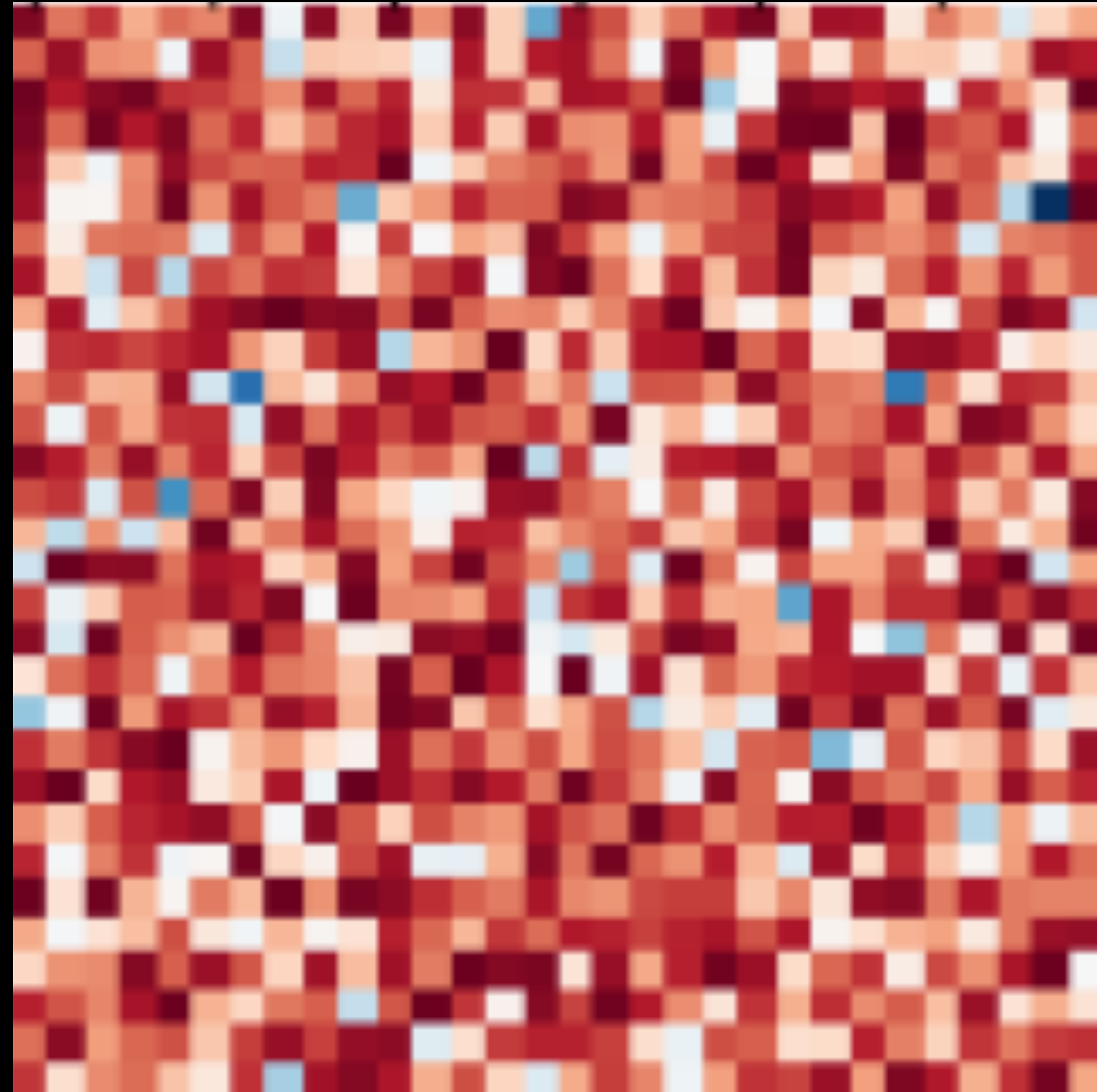
Context coupling

$$\left. \begin{aligned} J_{12}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_2 \\ J_{34}^{l+1} &= \frac{\partial}{\partial x_3^l} (f^{l+1}(X^l))_4 \end{aligned} \right\} A_{1234}^{l,l+1}$$



Depth-wise Coupling: Visualization of A

Llama-3 8B (Untrained)



Llama-3 8B

